

Exercise 20: Difference between JPA, Hibernate and Spring Data JPA

Aim:

To study and understand the difference between JPA, Hibernate, and Spring Data JPA.

JPA	Hibernate	Spring Data JPA
JPA (Java Persistence API) is a specification used for object-relational mapping in Java.	Hibernate is an ORM framework that implements the JPA specification.	Spring Data JPA is a Spring module that makes database operations easier by using JPA.
It only defines rules and interfaces.	It provides the actual implementation of those rules.	It uses JPA internally and reduces the amount of code developers need to write.
JPA cannot work by itself and needs an implementation like Hibernate.	Hibernate can work as a standalone ORM framework or as a JPA provider.	Spring Data JPA works with a JPA provider such as Hibernate.
It provides standard annotations like @Entity, @Id, and @Table.	It provides additional features like HQL, caching, lazy loading, and transaction support.	It provides ready-made repository interfaces such as JpaRepository and supports automatic query methods.
Main purpose: Define a standard way to interact with databases.	Main purpose: Map Java objects to database tables.	Main purpose: Simplify database access and reduce boilerplate code.

Result:

Thus, the differences between JPA, Hibernate, and Spring Data JPA were studied successfully. It was understood that JPA is a specification, Hibernate is its implementation, and Spring Data JPA simplifies database operations by providing repository support and reducing coding effort.